March 15, 2025

Submitted via Regulations.gov and Email

AI Action Plan
Attn: Faisal D'Souza
National Coordination Office
2415 Eisenhower Avenue
Alexandria, VA 22314

RE: "Request for Information on the Development of an Artificial Intelligence (AI) Action Plan," 90 Fed. Reg. 9,088 (February 6, 2025)

The Solar Energy Industries Association (SEIA) is the national trade association of the U.S. solar and storage industry. Our members promote the responsible development of distributed rooftop and utility-scale solar energy and storage projects. We are committed to working with federal agencies and other stakeholders to achieve this goal. As the national trade association for the U.S. solar and storage industry, which employs 280,000 Americans, SEIA represents over 1,200 organizations that manufacture, install, and support the development of solar and storage projects.

On behalf of our member companies, SEIA appreciates the opportunity to provide these comments on the Office of Science and Technology Policy's (OSTP) request for information, "Request for Information on the Development of an Artificial Intelligence (AI) Action Plan," 90 Fed. Reg. 9,088 (February 6, 2025) (RFI).[1]

I. Introduction

Advancing American energy dominance and security is a top priority for SEIA and the solar and storage industry. Solar energy is clean, abundant, and the United States has some of the richest solar resources in the world. It is an energy solution that provides reliable electricity, increases consumer choice, and helps consumers and business owners save money on their utility bills. Critically, solar energy components are increasingly made right here at home, positioning our nation as a geostrategic technology leader while reducing trade deficits.

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President Trump is correct that we must "sustain and enhance America's global AI dominance in order to promote human flourishing, economic competitiveness, and national security."[2] To assert our country's AI dominance, we must ensure there is enough energy to support the rapidly growing industry. Meeting the Administration's energy dominance and security goals will require an "all-of-the-above" energy strategy. We need every source of energy at our disposal to meet this historic demand, and solar and energy storage play a vital role.

II. Obstacles Created by Unprecedented Energy Demands

The RFI "seeks input on the highest priority policy actions that should be in the new AI Action Plan" on AI policy topics including "energy consumption and efficiency."[3] In order to reach the levels of energy production needed to match projected energy demand from the AI sector, the AI Action Plan should recognize the vital role of all energy sources, including solar and energy storage, in powering data centers.

With the surging adoption of digitization and AI technologies, the need for energy-intensive data centers has already skyrocketed and will continue to grow exponentially. Recent conservative estimates show that energy demand from U.S. data centers is expected to reach 606 terawatt-hours (TWh) by 2030, which is a drastic increase from the 147 TWh demand in 2023. Data centers currently make up 3.7% of the U.S. total energy demand, and this is projected to increase to 11.7% by 2030.[4]

After decades of flat power demand, the grid is now seeing significant increases in load growth.[5] This historic demand growth—the largest since World War II—presents great challenges at a moment in time that requires a comprehensive solution. The grid is aging. Old, uneconomic plants are retiring. And growing interconnection queues across the country have created long delays in bringing new sources of generation online.[6] While some data center developers have sought to co-locate their facilities with existing generation on the grid, such arrangements create

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[2] Executive Order 14179, "Removing Barriers to American Leadership in Artificial Intelligence" (January 23, 2025).

[3] 90 Fed. Reg. 9,088.

[4] Alastair Green et al., "How Data Centers and the Energy Sector can Sate AI's Hunger for Power" (September 17, 2024), *available at* https://www.mckinsey.com/industries/private-capital/our-insights/how-data-centers-and-the-energy-sector-can-sate-ais-hunger-for-power#/.

[5] John Wilson and Zach Zimmerman, "The Era of Flat Power Demand is Over" (December 2023), *available at* https://gridstrategiesllc.com/wp-content/uploads/2023/12/National-Load-Growth-Report-2023.pdf at 4.

[6] *Improvements to Generator Interconnection Procedures and Agreements*, Order No. 2023, 184 FERC ¶ 61,054, P 37 (2023).

potential grid reliability and resource adequacy concerns.[7] Ensuring significant new sources of reliable electricity to data centers and manufacturing facilities raises reliability, economic development, and national security concerns.

III. Natural Gas Generation Alone is Insufficient to Timely Meet Rising Demand

Electric utilities have been struggling to keep up with projected data center energy demands as generation and transmission projects are bottlenecked by various constraints and shortages. Natural gas combustion turbine plants, with potentially high fuel and pipeline infrastructure costs, are specifically designed to provide extra power during short periods of high demand and are not economic to run around the clock. Natural gas combined cycle plants, while designed to run at higher capacity factors, have recently seen sizable increases in costs and supply chain delays. Gas plants also must be located near existing major gas pipelines, which have their own development and construction challenges. Large gas plants can be the most challenging type of energy project to develop due to permitting, supply chain, labor, and other issues. Furthermore, all natural gas generators are exposed to potential fuel price increases and volatility, making their lifecycle costs unknowable and increasing commercial risks for offtakers.

These obstacles have recently led to large delays and cancellations for planned natural gas generation projects. For example, in February 2025, Engie withdrew two natural gas projects in Texas meant to help meet data center energy demand due to "equipment procurement constraints, among other factors."[8]

Both the largest builder of gas-fired generation in the U.S. over the past decade, NextEra, and the largest seller of gas turbines, GE Vernova, have recognized massive constraints for natural gas projects. NextEra's CEO recently stated on a January 2025 earnings call that:

> Gas-fired generation is moving forward but won't be available at scale until 2030, and then only in certain pockets of the U.S. In addition, gas-fired generation is more expensive than it's been, with costs having more than doubled over the last five years due to the limited supply of gas turbines, a constrained supply chain and much higher [engineering, procurement and construction] costs.[9]

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[7] *PJM Interconnection, L.L.C.*, Comm. Christie Concurring Statement, 189 FERC ¶ 61,078 (2024).

[8] *See* https://www.latitudemedia.com/news/engies-pulled-project-highlights-the-worsening-economics-of-gas/.

[9] *See* https://www.investor.nexteraenergy.com/~/media/Files/N/NEE-IR/news-and-events/events-and-presentations/2025/Final_Q4%202024%20Script.pdf.

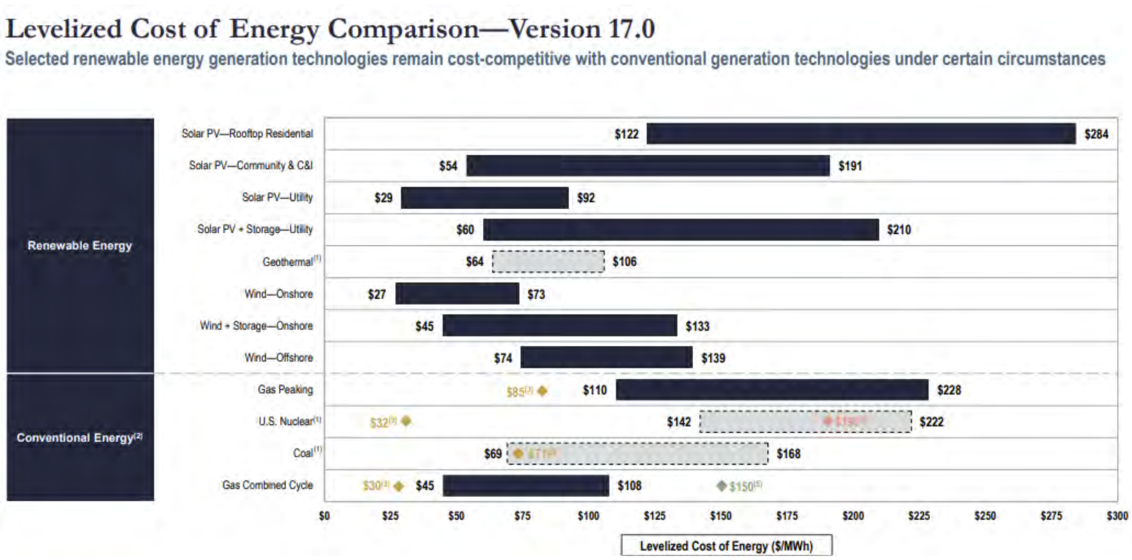
GE Vernova, which also partnered with Chevron to develop 4 gigawatts (GW) of gas power plants specifically for data centers, is now facing a backlog for new gas turbine deliveries stretching to 2029 or later.[10]

IV. The Role of Solar and Storage

Solar and storage is in many cases the best option for providing the massive amount of reliable power data centers need cheaply and quickly. For the reasons below, solar and storage have become the leading energy provider for data centers and other energy loads across the country.

A. Cost Effectiveness

Solar energy is one of the most cost-effective energy resources to build in the country. The levelized cost of electricity (LCOE) in 2024 for utility-scale solar ranges from $29/megawatt hour (MWh) to $92/MWh. The LCOE for natural gas peaker plants is 2.5 to 4 times more expensive with a range of $110/MWh to $228/MWh. The table below contains the 2024 LCOEs for each energy technology, which excludes tax incentives and variable fuel prices, and demonstrates the superior cost-effectiveness of utility-scale solar.[11]



Furthermore, solar energy offers more predictable long-term costs because it has a marginal fuel cost of zero. Financing involves the upfront cost of equipment, installation, and comparatively low maintenance costs.

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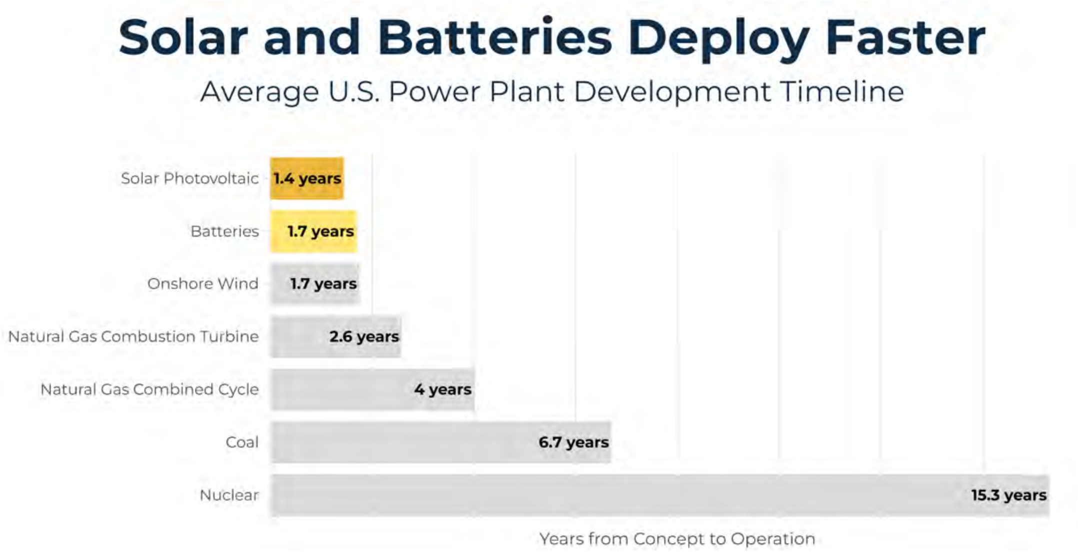
[10] *See* https://heatmap.news/ideas/natural-gas-turbine-crisis.
[11] "Levelized Cost of Energy+," Lazard, (June 2024), *available at* https://www.lazard.com/media/xemfey0k/lazards-lcoeplus-june-2024-_vf.pdf, at 9.

B. Construction Speed

Solar and energy storage are also the fastest energy resources to develop and deploy. Not only are they much simpler to engineer, but their supply chains are also more robust than natural gas (which currently faces a bottleneck in gas turbine blades). Speed to market is perhaps the most critical factor in meeting pressing data center energy demands, which cannot wait for other energy resources to begin production in 2030 or later.

Developing a utility-scale solar project typically takes only 1.4 years, and batteries only 1.7 years. The table below illustrates how these timelines are much faster than all other energy technologies.[12]



C. Reliability

Solar and storage have proven they can reliably provide the energy that data centers need. Solar not only generates electricity and supports the grid during normal operating conditions and periods, but also during times of increased grid risk such as extreme storms or periods of peak demand. Increasingly, utility-scale solar is being installed with battery storage, which can provide large amounts of stored energy to the grid when needed, bolstering its reliability and stability. Further studies have shown that high levels of solar and storage adoption can enhance the reliability of the U.S. electrical grid.[13]

[12] SEIA analysis of U.S. Energy Information Administration (EIA) Form 860M data for plants that have started reporting to EIA prior to seeking regulatory approval and plants which have reached operating status.
[13] Gregory Brinkman et al., "The North American Renewable Integration Study: A U.S. Perspective," NREL, (June 2021), *available at* https://www.nrel.gov/docs/fy21osti/79224.pdf.

Other energy sources have not fared as well. Nearly half of Texas' grid generation from gas and coal went down during a 2021 winter storm.[14] During a 2022 winter storm which battered the Midwest and Northeast, the Federal Energy Regulatory Commission (FERC) and the North American Electric Reliability Corp. (NERC) found that gas, coal, and oil-generated power accounted for 90% of the prolonged power outages experienced throughout the storm.[15] PJM, the largest grid operator in the U.S., issued a report for the same winter storm noting gas-fired generation accounted for 70% of unplanned outages, largely due to equipment failures from extreme cold and a lack of gas supply.[16] When Winter Storm Heather hit Texas in January 2024, the grid was supported with three times more solar and storage than in 2021, which played a crucial role in maintaining power during a record high peak demand.[17]

D. The U.S. is on a Path to Solar Energy Dominance

Overall, solar energy has become the proven leader in U.S. energy technologies. In 2024, the U.S. installed a record-breaking 50 GW of new solar capacity, the largest single year of new capacity added to the grid by any energy technology in over two decades. In addition to historic deployment, surging U.S. solar manufacturing emerged as a landmark economic story in 2024. Domestic solar module production tripled last year, and at full capacity, U.S. factories can now produce enough to meet nearly all demand for solar panels in the country. Solar cell manufacturing also resumed in 2024, strengthening America's energy supply chain and cementing its place as a solar powerhouse.[18]

Solar and energy storage have been the energy source of choice for "hyperscaler" builders of data centers. For example, in December 2024, Google announced a $20 billion investment in renewable energy, including solar and storage assets, by 2030 that will be "co-located" with their data centers. Google projects swift progress as its first co-located clean energy project is expected to start operating in 2026.[19]

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[14] Monika O'Shea et al., "Breaking Down the Texas Winter Blackouts: What Went Wrong?," Wood Mackenzie, (February 19, 2021), *available at* https://www.woodmac.com/news/editorial/Breaking-down-the-texas-winter-blackouts/full-report/.
[15] "Inquiry into Bulk-Power System Operations During December 2022 Winter Storm Elliott," FERC, NERC, and regional entities, (October 2023), *available at* https://www.ferc.gov/news-events/news/ferc-nerc-release-final-report-lessons-winter-storm-elliott.
[16] "Winter Storm Elliott Event Analysis and Recommendation Report," PJM, (July 17, 2023), *available at* 20230717-winter-storm-elliott-event-analysis-and-recommendation-report.ashx.
[17] *See* https://www.washingtonpost.com/climate-solutions/2024/01/14/texas-grid-winter-storm-power-outage/.
[18] "Solar Market Insight Report: Q4 2024," SEIA and Wood Mackenzie, (March 11, 2025), *available at* https://seia.org/research-resources/us-solar-market-insight/.
[19] *See* https://blog.google/inside-google/infrastructure/new-approach-to-data-center-and-clean-energy-growth/.

V. Conclusion

SEIA appreciates the efforts by OSTP to gather information to aid the development of the critical AI Action Plan. As described above, this is an important effort to achieve national AI dominance in order to promote human flourishing, economic competitiveness, and national security. It can only be achieved if we harness the power of all energy sources at our disposal. This includes the important role of solar and storage in effectively supporting the ongoing operations and future development of data centers. We look forward to continuing to work with you on this effort.

Thank you for the opportunity to provide comments. If you have any questions, please contact Ben Norris at (202) 556-2909 or bnorris@seia.org, or Alec Ward at award@seia.org.

Sincerely,

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